EX-5

Experiment on Packet Capture tool wireshark.

Aim:

Experiments on packet capture tool wireshark.

Packet sniffer:

→ Sniffs messages being sent/received from/by your computer.

→ Stores & displays the content of the various protocol fields in the messages.

→ Passive program

i) Never sends packets itself.

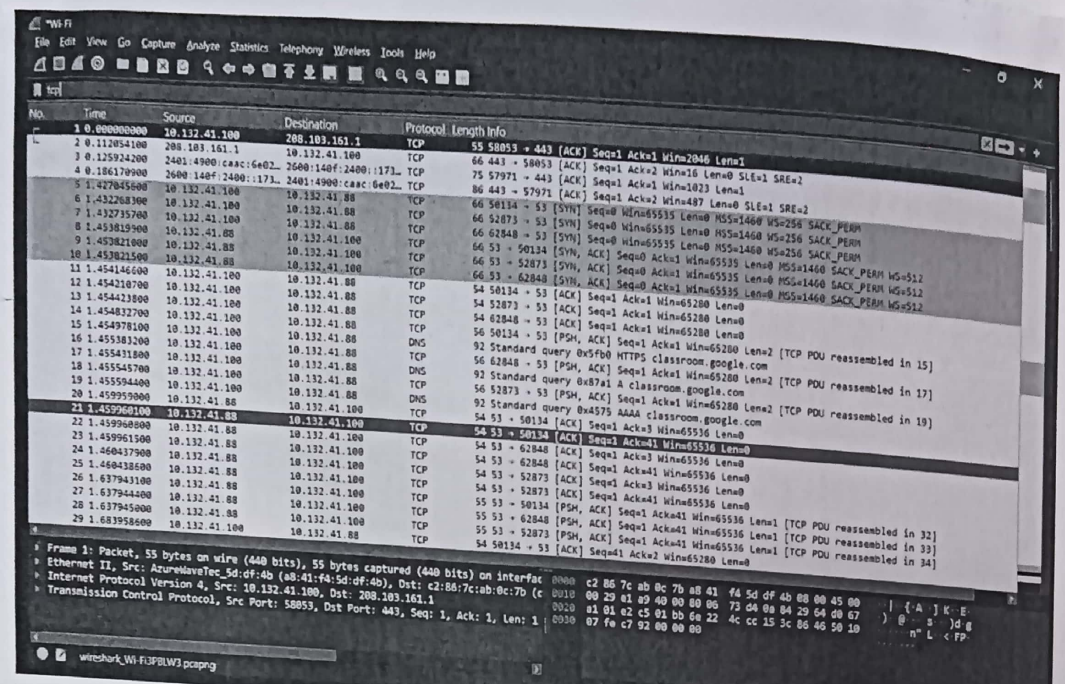
ii) No packets added to it.

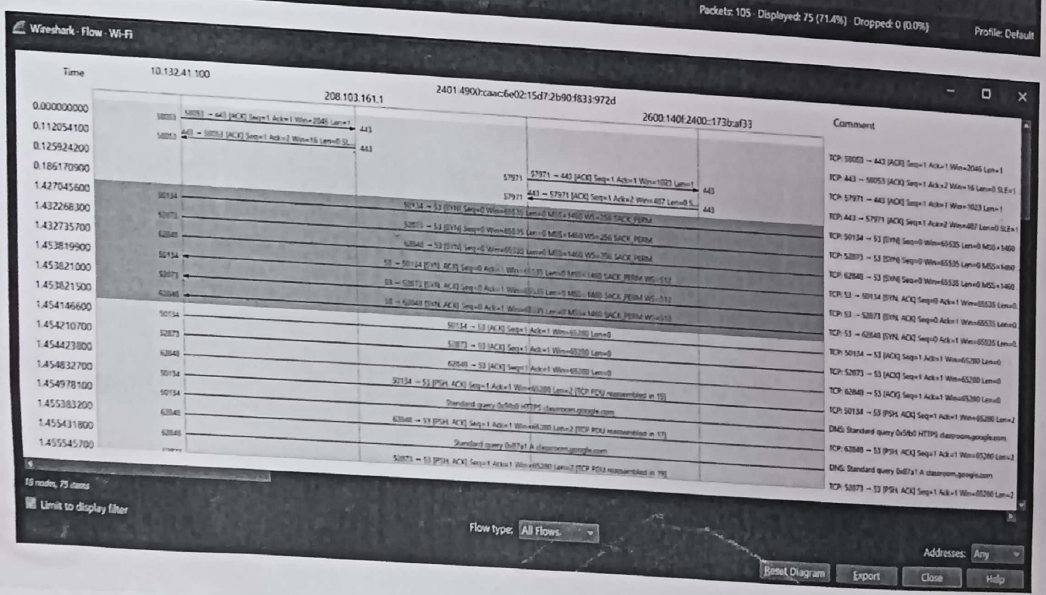
iii) Receives a copy of all packets (sent/received)

Packet Sniffer structure Diagnostic tool:

* Tcpdump - Eg: tcpdump -en x host 10.129 : 41.2
exe 3 out

* Wireshark - wireshark - r exe 3.out.





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Procedure:

→ Select the local area connection in wireshark

→ Go to Capture option.

→ Select capture & stop at automatically after 100 packets.

→ Then click start capture.

→ Save packets.

i) Create a filter to display only TCP/IP packets, inspect the packets & provide the flow graph

→ Go to search bar at the top of the tool

→ Search TCP Packets in search bar.

→ To see flow graph click statistics flow graph.

→ Save the packets.

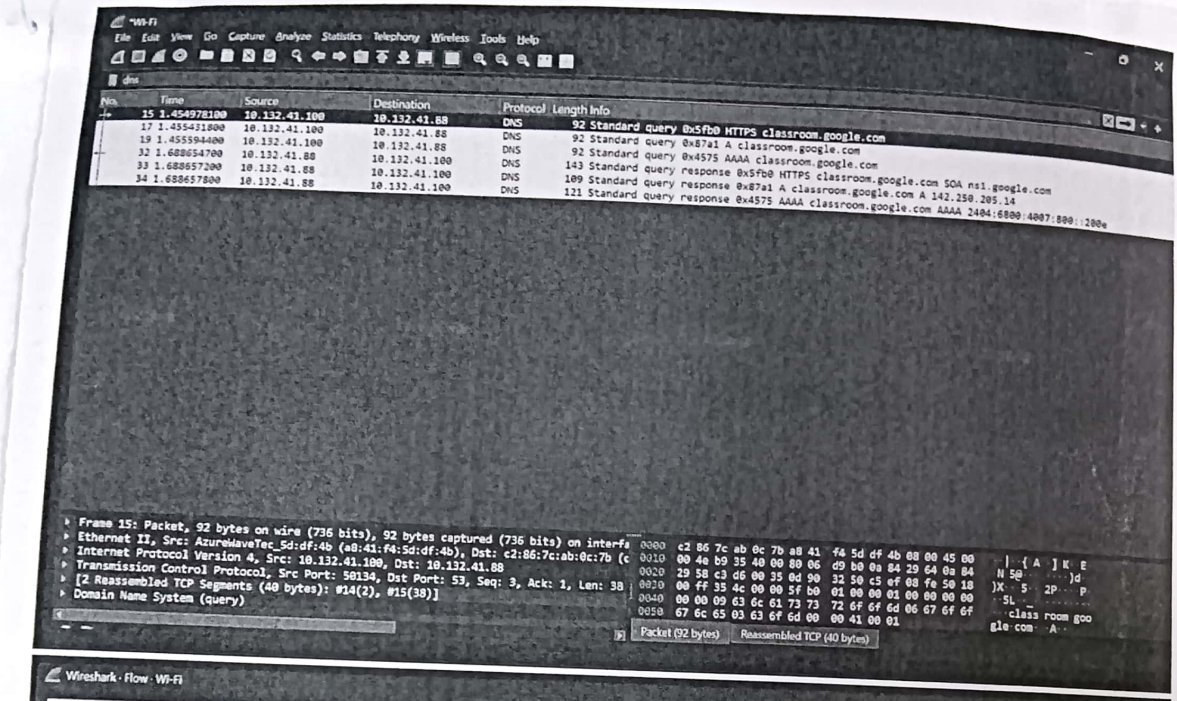
ii) Create a filter for ARP Packets

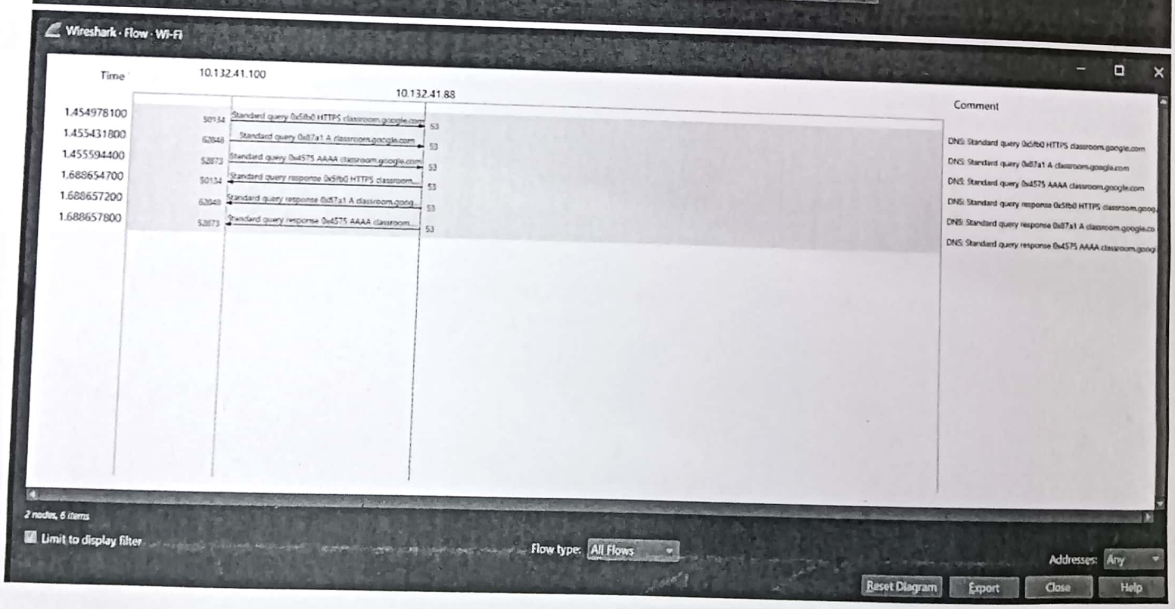
* Go to search option.

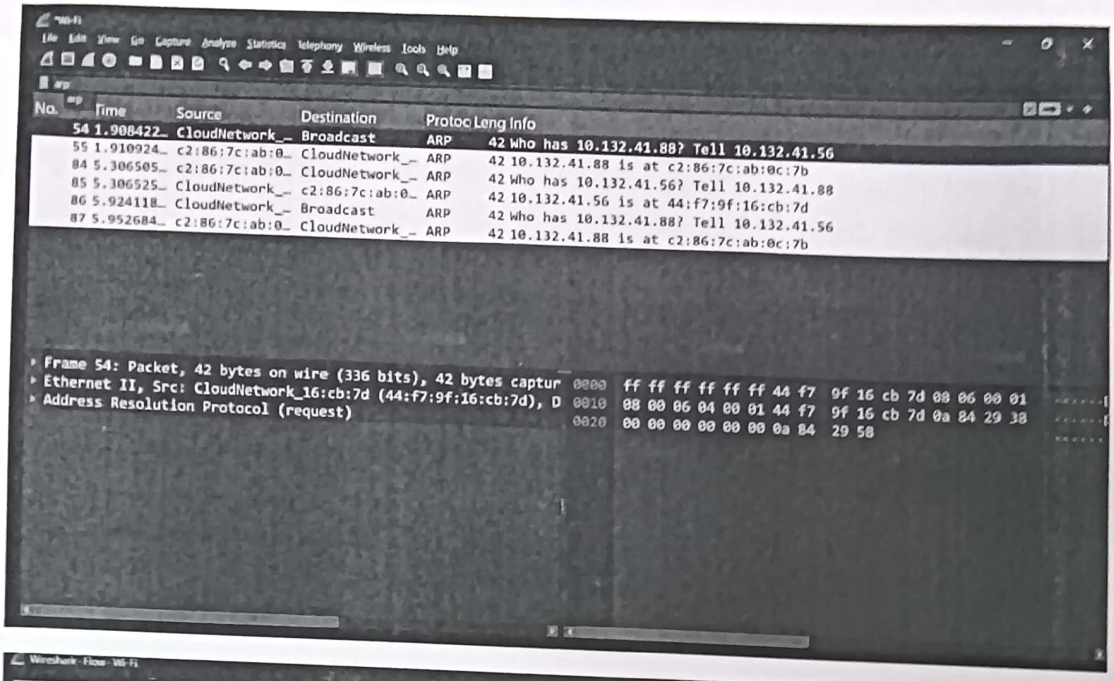
* Search ARP packets in search bar

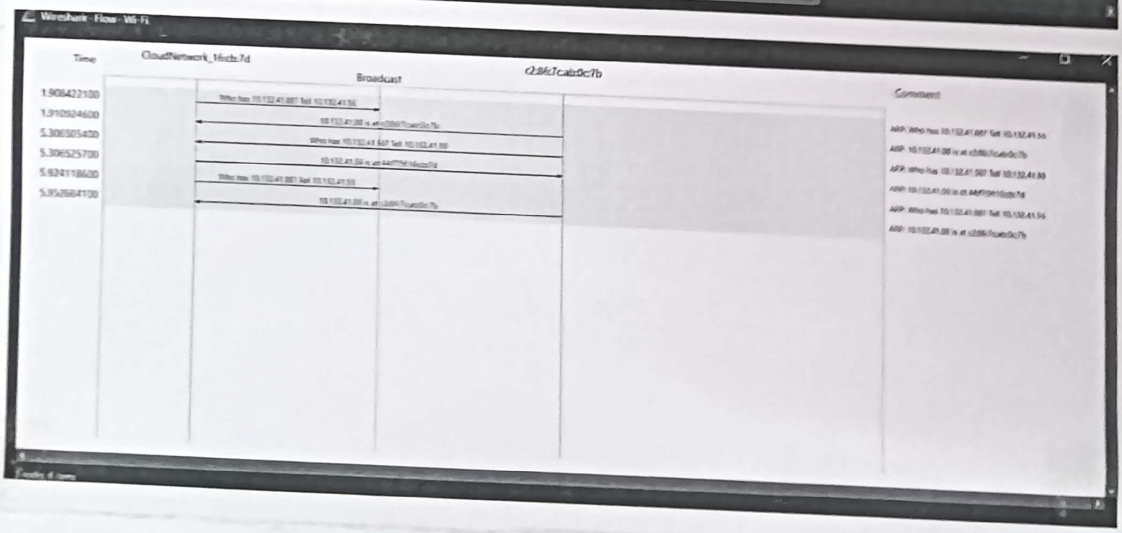
* Save the packets.

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*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
dns
No. Time Source Destination Protocol Length Info
15 1.454978100 10.132.41.100 10.132.41.88 DNS 92 Standard query 0x5fb0 HTTPS classroom.google.com
17 1.455431800 10.132.41.100 10.132.41.88 DNS 92 Standard query 0x87a1 A classroom.google.com
19 1.455594400 10.132.41.100 10.132.41.88 DNS 92 Standard query 0x4575 AAAA classroom.google.com
32 1.688654700 10.132.41.88 10.132.41.100 DNS 143 Standard query response 0x5fb0 HTTPS classroom.google.com SOA ns1.google.com
33 1.688657200 10.132.41.88 10.132.41.100 DNS 109 Standard query response 0x87a1 A classroom.google.com A 142.250.205.14
34 1.688657800 10.132.41.88 10.132.41.100 DNS 121 Standard query response 0x4575 AAAA classroom.google.com AAAA 2404:6800:4007:800::200e
Frame 15: Packet, 92 bytes on wire (736 bits), 92 bytes captured (736 bits) on interfa
Ethernet II, Src: AzureWaveTec_5d:df:4b (a8:41:f4:5d:df:4b), Dst: c2:86:7c:ab:0c:7b (c
Internet Protocol Version 4, Src: 10.132.41.100, Dst: 10.132.41.88
Transmission Control Protocol, Src Port: 50134, Dst Port: 53, Seq: 3, Ack: 1, Len: 38
[2 Reassembled TCP Segments (40 bytes): #14(2), #15(38)]
Domain Name System (query)
0000 c2 86 7c ab 0c 7b a8 41 f4 5d df 4b 08 00 45 00
0010 00 4e b9 35 40 00 80 06 d9 b0 0a 84 29 64 0a 84
0020 29 58 c3 d6 00 35 0d 90 32 50 c5 ef 08 fe 50 18
0030 00 ff 35 4c 00 00 5f b0 01 00 00 01 00 00 00 00
0040 00 00 09 63 6c 61 73 73 72 6f 6f 6d 06 67 6f 6f
0050 67 6c 65 03 63 6f 6d 00 00 41 00 01
Packet (92 bytes)
Reassembled TCP (40 bytes)


Wireshark · Flow · Wi-Fi
Time
10.132.41.100
10.132.41.88
Comment
1.454978100
1.455431800
1.455594400
1.688654700
1.688657200
1.688657800
Standard query 0x5fb0 HTTPS classroom.google.com
Standard query 0x87a1 A classroom.google.com
Standard query 0x4575 AAAA classroom.google.com
Standard query response 0x5fb0 HTTPS classroom...
Standard query response 0x87a1 A classroom.goog...
Standard query response 0x4575 AAAA classroom...
2 nodes, 6 items
Limit to display filter
Flow type: All Flows
Addresses: Any
Reset Diagram
Export
Close
Help



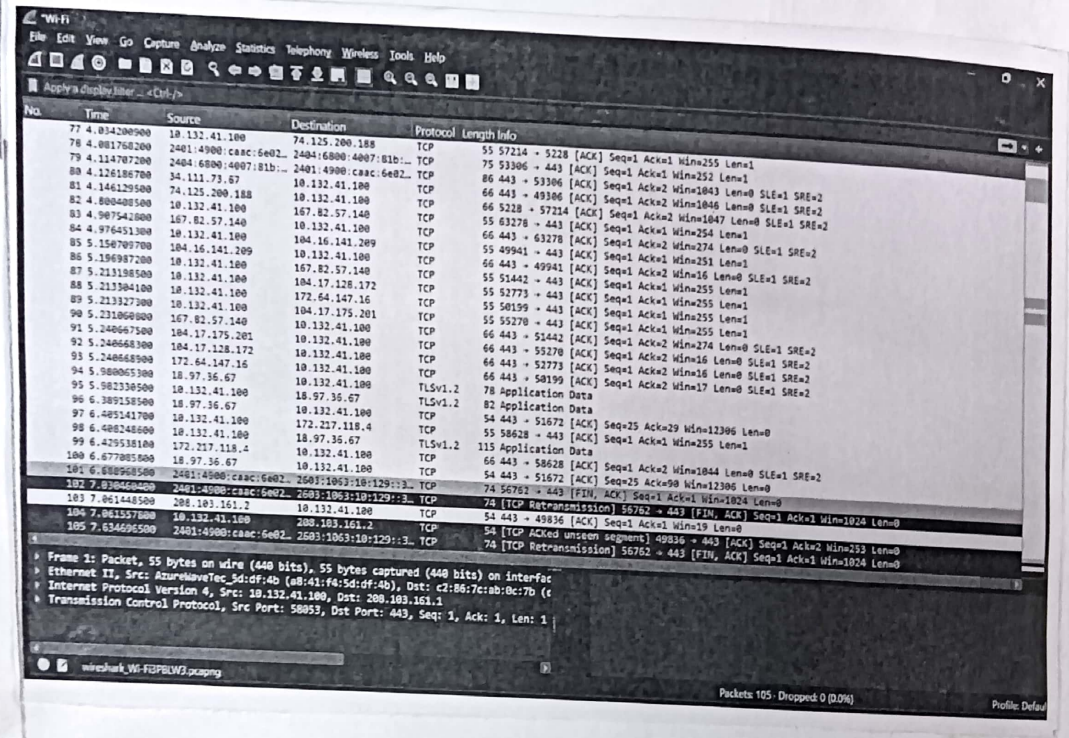


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iii) Creating a filter to display DNS packets & provide the flow graph.

iv) Creating a filter for ICMP & DHCP pack & inspect & show the packets.

v) Create filter for ICMP & DHCP packets & inspect & show the packets.



Result:-

Wireshark helps us in analysing & understanding our network traffic & packet flow through various filters & flow graphs.

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